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Professional Cloud Architect: Day 2 Polls



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Load Balancing



Load Balancing

Your team needs to deploy a legacy financial application on Google Cloud. The application communicates over TCP and relies on direct access to a local filesystem to maintain data integrity. It cannot be scaled horizontally because it does not have synchronization and accessing the file system concurrently causes problems that cannot be resolved easily. Although brief downtime is acceptable during failover, the application must remain highly available to support continuous business operations. How should you architect this deployment?

- A. Deploy the application on a managed instance group across multiple zones, use Cloud Filestore for shared storage, and place an HTTP(S) load balancer in front to distribute traffic.
- B. Deploy the application on a managed instance group across multiple zones, use Cloud Filestore for shared storage, and use a network load balancer to balance traffic.
- C. Create an unmanaged instance group with one active and one standby VM in different zones, attach a regional persistent disk for storage, and place an HTTP(S) load balancer in front to route requests.
- D. Create an unmanaged instance group with an active VM and a standby VM in separate zones, use a regional persistent disk for data storage, and place a network load balancer in front to route client connections.



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Load Balancing

A media company is building a global video streaming platform on Google Cloud using dozens of microservices. They require a CI/CD pipeline for frequent, independent updates using immutable artifacts. The architecture must provide low-latency access to users in both North America and Asia via a single global entry point, while keeping core backend services private from the internet. Which set of Google Cloud services is most suitable for this architecture?

- A. Artifact Registry to store container images, Google Kubernetes Engine (GKE) clusters in an Asian and North American region, and a Global External HTTP/S Load Balancer.
- B. App Engine to host the services in multiple regions, Cloud Spanner for a global database, and a regional Internal TCP/UDP Load Balancer.
- C. Cloud Storage to store application binaries, Managed Instance Groups (MIGs) in each region, and a Regional External Network Load Balancer for each region.
- D. Cloud Functions for the microservice logic and a Global External HTTP/S Load Balancer configured with Serverless Network Endpoint Groups (NEGs).



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Load Balancing

An international e-commerce company has deployed its containerized shopping cart API on Cloud Run in two regions: europe-west1 and australia-southeast1. They need to provide a single global endpoint (api.shopping.com) for their frontend applications that automatically routes customers to the closest healthy region, ensuring a fast and resilient shopping experience.

What is the recommended Google Cloud native approach to achieve this?

- A. Create a **Regional External HTTP/S Load Balancer** in both regions. Use Cloud DNS to create weighted A records to distribute traffic between the two load balancer IPs.
- B. For each regional Cloud Run service, create a **Serverless Network Endpoint Group (NEG)**. Create a single **Global External HTTP/S Load Balancer** and configure its backend service to use these two Serverless NEG.
- C. Use **API Gateway** in front of each Cloud Run service. Configure the two API Gateway instances under a single custom domain in Cloud DNS to handle routing.
- D. Create a **Global External HTTP/S Load Balancer**. Manually add the default .run.app URLs of the two Cloud Run services as **Internet Network Endpoint Group** backends to the load balancer.



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Networking



Networking

A financial services company is extending its on-premises data center into Google Cloud to create a hybrid environment. They plan to run application servers on Google Cloud that need to communicate directly with a database server remaining on-premises, using only private, internal IP addresses (RFC 1918).

The company requires a networking solution that provides continuous connectivity even when an existing link is down.

Which combination of services should they implement to achieve this?

- A. Use a **Global External HTTP/S Load Balancer** with an on-premises server as an Internet NEG backend, and a second load balancer for path diversity.
- B. Establish a **Dedicated Interconnect** for a private physical path. For path diversity, configure **Direct Peering** to Google's network edge.
- C. Set up a **Shared VPC** to link the on-premises data center and the Google Cloud VPC, and use **Cloud Router** to manage routes.
- D. Establish a **Dedicated Interconnect** for a high-bandwidth, private physical path. Concurrently, configure a **Cloud VPN** tunnel to provide an encrypted path over the public internet.



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Networking

A stock photography company, "PixelPerfect," hosts its library of high-resolution images in a Cloud Storage bucket. Their website, running on a fixed set of Compute Engine instances, serves these images to a global user base of graphic designers.

Customers are reporting that downloading the most popular and featured images of the week is frequently slow and sometimes times out, especially during peak business hours in their respective regions. You need to accelerate the delivery of these popular images and improve the reliability of the download experience for all users.

What should you do?

- A. Write a script that syncs the Cloud Storage bucket to a Persistent Disk attached to each Compute Engine VM. Modify the application to serve the images directly from the VMs' local disks.
- B. Provision a **Memorystore for Redis** instance. For popular images, load the raw image data into Redis as byte strings. Update the application to check Redis first before fetching from Cloud Storage.
- C. Create a **Bigtable** table to store the image data. Rewrite the application to stream the images from Bigtable rows to reduce the load on Cloud Storage for popular files.
- D. Set up a **Global External HTTP/S Load Balancer**. Configure one of its backend services to be the **Cloud Storage bucket** containing the images. Ensure that **Cloud CDN** is enabled for this backend service.



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A retail company is building a hybrid application where a new Cloud Function on GCP is used for real-time inventory checks. This function needs to query a high-performance Redis cache that resides in their on-premises data center.

The company's Google Cloud VPC is already connected to their on-premises network via a stable Cloud VPN tunnel. The Cloud Function must be able to send requests to the private IP address of the on-premises Redis cache.

What must be configured in Google Cloud to enable this communication path?

- A. Modify the Cloud VPN tunnel's firewall rules to explicitly allow traffic originating from the public IP ranges of Google Cloud Functions.
- B. Assign a public static IP address to the on-premises Redis cache using a NAT gateway and have the Cloud Function connect to it over the public internet.
- C. Configure Private Google Access for the on-premises network, allowing on-premises hosts to communicate privately with Google services like Cloud Functions.
- D. Create a Serverless VPC Access connector. Configure the Cloud Function to route its traffic through this connector into the VPC, allowing the traffic to then traverse the Cloud VPN tunnel.



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Networking

A healthcare research institute stores sensitive patient data in Google Cloud Storage buckets within a dedicated GCP project. Researchers access this data for analysis from Compute Engine VMs inside the same project's Virtual Private Cloud (VPC).

The security team must enforce a strict security perimeter to prevent data exfiltration. They need to block any attempt by a compromised VM or a malicious user to copy data from the project's secure buckets to a public bucket or any other unauthorized Google Cloud location, while still allowing the VMs to access the required Cloud Storage APIs for their work.

What is the most effective Google Cloud solution to meet these requirements?

- A. Create a custom IAM role that only grants `storage.objects.get` permission. Assign this role to researchers and remove all broader storage roles to enforce least privilege.
- B. Enforce the use of Customer-Managed Encryption Keys (CMEK) for all Cloud Storage buckets. Only grant key decryption permissions to authorized service accounts.
- C. Define a VPC Service Controls perimeter that includes the project and the required Google APIs (e.g., Cloud Storage API). Ensure Private Google Access is enabled on the VPC's subnets.
- D. Configure the project's VPC firewall rules to deny all egress traffic to the internet. Enable Private Google Access on the subnet to allow VMs to continue reaching Google APIs.



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Networking

A financial company is deploying applications on a private Google Kubernetes Engine (GKE) cluster to maximize security. The GKE nodes in this cluster have been provisioned with no external IP addresses and cannot be reached directly from the internet.

However, to launch new applications, these nodes must be able to pull container images from public registries like Docker Hub, which are located on the internet. You need to enable this outbound connectivity for the GKE nodes without assigning them public IPs.

What is the recommended and most scalable Google Cloud solution?

- A. Enable Private Google Access on the GKE cluster's subnet to allow the nodes to reach public endpoints on the internet.
- B. Deploy a Compute Engine VM with a public IP to act as a forward proxy server. Configure the GKE nodes to route all outbound traffic through this proxy.
- C. Provision a Cloud NAT gateway for the VPC network and configure it to apply to the subnet where the private GKE nodes reside.
- D. Configure an Internal TCP/UDP Load Balancer within the VPC and create a firewall rule that allows it to forward traffic to the internet.



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Storage Services



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A film studio is creating a central repository on Google Cloud for its global post-production teams. The first step is to migrate their historical archive of 800 TB of raw 8K video footage from their on-premises storage systems.

Furthermore, their active film sets generate approximately 15 TB of new footage *each day*, which must also be uploaded promptly to the cloud repository for editing. The studio's existing internet connection is a 1 Gbps line.

What data transfer strategy should the studio adopt to handle both the initial archive migration and the ongoing daily uploads efficiently?

- A. Use Storage Transfer Service to configure a transfer job from the on-premises storage to Cloud Storage. Run this job continuously over the existing 1 Gbps internet link until both the archive and daily footage are uploaded.
- B. For the initial 800 TB archive, use Transfer Appliance to ship the data physically to Google. For the ongoing 15 TB daily uploads, establish a Dedicated Interconnect of at least 10 Gbps.
- C. Use Transfer Appliance for the initial 800 TB archive. For the daily uploads, set up multiple Cloud VPN tunnels over the existing 1 Gbps internet connection to aggregate bandwidth.
- D. For the initial archive, provision a 10 Gbps Dedicated Interconnect and transfer the 800 TB over this link. For the daily uploads, use Transfer Appliance, ordering a new one each day.



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A logistics company is building a data lake on Google Cloud to ingest real-time JSON telemetry data from thousands of IoT sensors on its trucks. The structure of the JSON payload can change unexpectedly when sensor firmware is updated.

A key business requirement is that the raw, unaltered JSON messages from every sensor must be preserved indefinitely in their original format. This ensures that data can be reprocessed if a downstream analytics pipeline fails due to a schema change.

What is the most appropriate initial step for the data ingestion architecture to meet this requirement?

- A. Stream the JSON data directly into a Dataflow pipeline that validates and transforms the data. Store the resulting structured data in BigQuery.
- B. Create a Cloud Storage bucket to serve as the landing zone. Ingest the raw JSON messages and store each one as an individual, immutable object in this bucket before any processing occurs.
- C. Ingest the JSON messages directly into a Cloud Bigtable table, with each message's fields stored in separate columns. Design the processing pipelines to read from Bigtable.
- D. Publish all incoming JSON messages to a Pub/Sub topic. Set the message retention duration on the topic to its maximum period to allow for reprocessing from the topic itself.



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Storage Solutions

You are designing a backend service for an e-commerce platform that will handle and store transactional data from mobile and web users located worldwide. With the platform's global launch, you expect to handle petabytes of transaction data, and the business team needs the ability to run SQL queries for analysis. You need to create a data store that is highly available, scalable, and capable of efficiently handling massive volumes of data. What should you do?

- A. Use Cloud Spanner to create a globally distributed, scalable, and highly available relational database that supports SQL queries.
- B. Use Firestore in Datastore mode to store transactional data and run SQL queries through third-party tools.
- C. Set up a Cloud SQL instance with regional replication to handle the large volume of global transactions.
- D. Deploy a MySQL database on Compute Engine VMs and configure replication across regions to handle the data load.



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Storage Solutions

You are designing a system to ingest and store data from millions of IoT sensors around the world. The data will be sent continuously at high throughput, and the business team needs to query and analyze this data based on the time it was generated. You want to ensure that the system can scale efficiently and handle real-time queries. What should you do?

- A. Ingest the data into BigQuery, and partition the tables by timestamp for querying.
- B. Ingest the data into Cloud SQL and index it by timestamp for efficient queries.
- C. Ingest the data into Bigtable, and create a row key based on the event timestamp for optimized querying and scaling.
- D. Ingest the data into Firestore, and organize the data based on event timestamps for real-time analysis.



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- A. Use a single Cloud SQL instance and configure automated backups for disaster recovery.
- B. Use Cloud SQL configured for high availability with failover replicas in different zones.
- C. Deploy a Cloud SQL instance in a single zone and set up replication using Cloud Spanner.
- D. Use Bigtable with replication in the same zone to ensure availability and performance.



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Identity and Access Management



IAM

A financial corporation uses two separate Google Cloud projects: `finance-data-project` for storing sensitive quarterly reports in a Cloud Storage bucket, and `analytics-pipeline-project` for running a data processing pipeline on Compute Engine.

The pipeline running in `analytics-pipeline-project` needs programmatic, read-only access to the reports in the Cloud Storage bucket located in `finance-data-project`. You must provide this access following the principle of least privilege, granting only the absolute minimum permissions required.

What is the most secure and appropriate method to grant this access?

- A. Move the Compute Engine instance from the `analytics-pipeline-project` into the `finance-data-project` so they are in the same project and can communicate directly.
- B. In the `finance-data-project`, grant the Project Editor role to the default Compute Engine service account from the `analytics-pipeline-project`.
- C. In the `finance-data-project`, create a new **service account**. Grant this service account the **Storage Object Viewer** role on the specific bucket. Then, grant the Compute Engine service account in the `analytics-pipeline-project` the permission to impersonate this new service account.
- D. Make the Cloud Storage bucket public so the analytics pipeline can read the data, but use a firewall rule to restrict access to only the IP address of the Compute Engine instance.



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IAM

A European bank is migrating to Google Cloud and must adhere to strict GDPR data sovereignty rules. Their corporate policy dictates that all cloud resources and stored data must physically reside within European Union data centers.

What is the most effective and scalable Google Cloud feature to enforce this policy?

- A. Submit requests to Google Cloud Support to reduce the resource quotas for every service to zero in all non-EU regions.
- B. Define an Organization Policy with the `gcp.resourceLocations` constraint. In the policy, specify the list of allowed Google Cloud locations to include only the EU regions.
- C. Create a Cloud Audit Logs sink that exports all resource creation logs to BigQuery. Set up a scheduled query to run every hour to report on any resources created outside the EU. Set up a script to immediately shutdown those resources
- D. Configure the organization's VPC firewall rules to deny all traffic that originates from or is destined for IP ranges associated with non-EU regions.



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IAM

You are deploying a new virtual machine (VM) on Google Cloud to run a specific batch processing job. The VM needs to read data from a Cloud Storage bucket and write results to a Cloud SQL instance. You want to configure its access to these services by following the principle of least privilege and Google Cloud's recommended security practices.

What should you do?

- A. Modify the project's default Compute Engine service account by adding the Cloud SQL Client and Storage Object Admin roles to it. Attach this default service account to the new VM.
- B. Create a new, dedicated service account for the VM. To ensure it has sufficient permissions, grant it the project-level Editor role. Attach this service account to the VM.
- C. Create a new, dedicated service account and grant it the necessary roles (Cloud SQL Client, Storage Object Viewer). Generate a JSON key for this service account, securely copy it to the VM, and configure the application to use this key for authentication.
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Security



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A fintech company's fraud detection team needs to analyze a live stream of financial transactions to identify suspicious patterns. The raw transaction events, by an API, contain sensitive data, including full credit card numbers.

For compliance with payment card industry regulations (PCI DSS), the raw credit card numbers must not be stored at all on the Google Cloud

What is the most appropriate and secure Google Cloud solution?

- A. Create a streaming Dataflow job that reads from the API. Within the pipeline, use a Cloud Data Loss Prevention (DLP) transformation to inspect the data and tokenize the credit card numbers. Write the transformed, de-identified records to BigQuery.
- B. Stream the raw data directly into a staging table in BigQuery. Use BigQuery's column-level security and IAM policies to prevent analysts from viewing the column that contains the credit card numbers.
- C. Create a streaming Dataflow job that reads from the API and pushes the data to Pub/Sub. Write a Cloud Function that triggers on new Pub/Sub messages. The function should write transaction records containing credit card numbers to a highly restricted, encrypted Cloud Storage bucket, and write all other records to BigQuery.
- D. Create a Dataflow job that uses a custom Regular Expression (Regex) to find and replace number patterns that look like credit card numbers. Load the result into BigQuery.



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A financial services company is landing sensitive customer transaction data in a Cloud Storage bucket before loading it into BigQuery for analytics. For compliance with industry regulations, they must manage and be able to rotate the encryption keys used for this data at rest.

A critical requirement is that BigQuery must be able to transparently read the data from the Cloud Storage bucket without a separate, manual decryption step in the pipeline.

What is the recommended Google Cloud practice to meet these requirements?

- A. Before uploading, use an application to encrypt each file with a key stored in Cloud KMS. The BigQuery load job must then be configured to call the KMS API to decrypt each file.
- B. Create a cryptographic key in Cloud KMS. Configure the Cloud Storage bucket's default encryption settings to use this Customer-Managed Encryption Key (CMEK). Grant the BigQuery service account the necessary role on the KMS key.
- C. Generate your own AES-256 key on-premises. When uploading files to Cloud Storage, provide this key with the request to use the Customer-Supplied Encryption Key (CSEK) feature.
- D. Store the encryption key in Secret Manager. Write a Cloud Function that triggers on new file uploads, reads the key from Secret Manager, and re-writes the file with the encryption applied.



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Observability



Observability

A company runs critical nightly batch processing jobs on a Google Kubernetes Engine (GKE) cluster. The runtime for these jobs has been increasing, and they are now at risk of violating their Service Level Agreement (SLA).

The cluster was initially provisioned without any integrated observability suite, so there is no historical performance data available. You need to collect performance metrics (like CPU and memory utilization) from the running pods to diagnose the bottleneck. The solution must not require re-deploying the application or disrupting the currently running jobs.

What is the most direct and least disruptive way to accomplish this?

- A. Execute a `gcloud container clusters update` command on the existing cluster to enable Cloud Operations for GKE. Then, use Cloud Monitoring to analyze the performance metrics of the pods.
- B. Cordon and drain the existing GKE nodes one by one. Replace them with new nodes from a new node pool that has Cloud Operations for GKE enabled, allowing the jobs to be rescheduled onto new, monitored nodes.
- C. Write a script that repeatedly uses `kubectl top` and `kubectl logs` to manually sample performance data from the pods while the job is running. Store the output in a text file for later analysis.
- D. Deploy a third-party monitoring agent, like Prometheus, to each node in the cluster using a Kubernetes DaemonSet. Configure the agent to collect metrics and send them to Cloud Monitoring.



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Your company's security team needs to meet SOX compliance requirements by retaining all Admin Activity audit logs for a period of five years. These logs track administrative changes to your Google Cloud resources and must be stored in a secure, immutable, and cost-effective manner for potential future audits.

The solution must capture only these specific audit logs from your entire Google Cloud Organization.

What is the most appropriate and direct way to configure this?

- A. Enable Data Access audit logs for all services to ensure maximum visibility. Create a sink to export these high-volume logs to a BigQuery table for detailed analysis. Get BigQuery partitions to expire in 5 years
- B. Upgrade the default log bucket to use Log Analytics. Run a scheduled query every day to view the Admin Activity logs from the past 24 hours to ensure they are captured. Make sure you can query 5 years worth of data
- C. Create a log sink at the organization level. Configure the sink's filter to select only Admin Activity audit logs and set the destination to a Cloud Storage bucket configured with a 5-year retention lock.
- D. Create a log-based alert in Cloud Monitoring that triggers whenever an Admin Activity log is generated. Configure the alert to send a notification to a Pub/Sub topic for real-time tracking and storage. Store messages for 5 years



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